

NBA True Scoring Value Model

Project goal

Build an advanced analytics model that estimates individual NBA players true scoring value/impact that traditional box score statistics like PPG and FG%, as well as advanced metrics such as usg% and ts% are incapable of doing.

Tech Stack

Python, Pandas, NumPy, scikit-learn, Plotly, Streamlit

Project Structure

data/ → raw and cleaned datasets models/ → scoring impact model files app/ → Streamlit web app
notebooks/ → EDA and analysis

EDA Findings

- High PPG does not equal high efficiency. Elite scorers like SGA and Jokić score fewer points than Luka but shoot far more efficiently.
- As usage rate increases, efficiency regresses toward the league mean. Elite players who maintain high efficiency at high usage are beating the expected tradeoff.
- Scoring is right skewed (most players average 5-15 PPG), making 20+ PPG scorers statistical outliers.
- Scoring volume (PTS) and efficiency (TS_PCT) have a weak correlation ($r = 0.25$), confirming that PPG alone is a misleading measure of scoring quality.